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Single-Agent Reinforcement Learning Model for Adaptive Traffic Signal Control in Urban Corridors

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ABSTRACT

For multi-intersection control, existing research mainly adopts multiagent frameworks to tackle scalability issues. However, the traffic signal control (TSC) problem necessitates a single-agent framework, as a single control center monitors traffic conditions across all roads in the study area and coordinates the control of all intersections. This work proposes a novel single-agent RL-based urban corridor ATSC model: It abandons complex multiagent coordination and uses a single agent to centrally orchestrate signal timings across multiple intersections. Notably, the model is highly applicable to real-world settings. It defines state and reward functions based on a queue length metric—one that correlates with congestion and can be reliably estimated using probe vehicle data. Since probe vehicle data have become highly prevalent, this feature enables rapid, large-scale deployment. The single-agent framework primarily relies on a unique design of state, action, and reward. To facilitate learning and manage congestion, both state and reward functions are defined based on queue length, with actions designed to modulate queue dynamics. The queue length definition used in this study deviates slightly from conventional definitions but is closely correlated with congestion states. The method was comprehensively evaluated using the SUMO simulation platform under various traffic patterns. Experimental results show that the PPO algorithm demonstrates significantly faster learning than the DQN algorithm. The model effectively alleviates urban corridor congestion through coordinated multi-intersection control: During the simulation period of the entire scenario, the queue length did not exceed 50 vehicles, and instances where it exceeded 30 vehicles were relatively rare. Compared with the baseline scenario where queue lengths exceeded 150 vehicles, the proposed method significantly reduces road congestion. The work in this paper demonstrates the feasibility of controlling multiple intersections under a single-agent framework, and the control scope will be further expanded in the future.

1 | Introduction

Traffic congestion significantly impacts urban quality of life, traffic safety, environmental conditions, and economic performance, thereby emerging as a critical issue in numerous cities worldwide. One of the primary contributors to urban road congestion is queuing at intersections. Traffic signal control (TSC) systems play a pivotal role in managing traffic flow at these signalized intersections and hold substantial potential for mitigating traffic congestion and enhancing travel efficiency.

Traditionally, traffic engineers have sought to optimize traffic signal timings through the formulation of optimization models based on certain traffic dynamics assumptions [1]. While these

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models make TSC problems more tractable, they often deviate from real-world conditions. Consequently, they fail to encompass the complex and stochastic scenarios encountered in urban traffic, thereby diminishing their effectiveness in swiftly adapting to dynamic traffic demands [2].

With the rapid advancements in artificial intelligence, a novel and promising learning-based approach, known as reinforcement learning (RL), has emerged to address the limitations inherent in traditional model-based control strategies [3]. Through ongoing research efforts, RL has evolved into a widely investigated technology for the development of smart TSC systems [4, 5]. It has demonstrated considerable suitability for managing complex and dynamic traffic environments [1, 6–8].

When it comes to the control of multiple intersections, researchers generally believe that due to the rapid expansion of the scale of the state space and action space, it is extremely challenging to achieve effective control with a single-agent framework. As a result, the focus has shifted toward multiagent control systems, which have shown remarkable potential in terms of scalability [7, 9, 10].

However, adopting multiagent systems is merely a second-best solution for addressing scalability. TSC is essentially centrally controlled by a single control center. This center is capable of observing the traffic conditions of all roads within the study area and coordinating the control of all intersections. This is fundamentally different from the task coordination and control of drones or automated guided vehicles (AGVs). Moreover, both local observation ranges and interagent messages are artificially defined, introducing unnecessary complexity not present in real-world TSC scenarios.

Currently, probe vehicle technology is a highly competitive traffic monitoring technology for TSC systems [11]. Firstly, probe vehicle data cover most urban roads, eliminating the need for additional equipment to gather traffic information. By utilizing vehicles from ride-hailing services and users of navigation systems, it is possible to collect real-time trajectories for 10%–30% of the vehicles on the road network. Secondly, probe vehicle technology can estimate queue length, a metric that current sensors, such as induction loops, find challenging to capture [12]. Queue length is a well-recognized indicator that effectively describes the congestion level on urban roads. Therefore, the compatibility of TSC methods with probe vehicle systems will contribute to their rapid popularization.

In this work, we propose a single-agent RL-based urban corridor TSC model compatible with probe vehicle technology. The contributions of this work are summarized as follows:

1. We propose a single-agent RL model for urban corridor TSC. This model uniquely employs a single agent to orchestrate signal timing across multiple intersections, presenting a paradigm shift from conventional multiagent RL approaches. By eliminating interagent coordination overhead and maintaining a unified policy optimization process, our model achieves remarkable simplicity.

2. In RL environment design, both the state and reward functions are based on queue length, and the action effectively manages queue length, thereby accelerating policy learning. The queue length definition adopted in this study differs slightly from conventional definitions. This definition is closely correlated with congestion states and, more importantly, allows for reliable estimation using link travel time data from probe vehicles, alongside automatic license plate recognition, connected vehicles, and other technologies. This characteristic gives the proposed method the potential to be deployed on most urban roads.

3. A simulation experiment was established to validate the proposed model. First, two patterns of origin–destination (OD) matrix scenarios were tested. In the first pattern, one direction of flow is set to be dominant to simulate a corridor connecting suburban and central areas of a city. In the second pattern, the two directions of flow are set to be equal, simulating a corridor located within the central area of a city. Second, we investigate proximal policy optimization (PPO) and deep Q-network (DQN) algorithms to evaluate their learning efficiency and the congestion control efficiency.

The remainder of this paper is organized as follows. Section 2 describes the related works. Section 3 proposes the RL-based TSC model. Section 4 presents the experiment design. Section 5 conducts the experiments and discussion. Finally, Section 6 concludes this work.

2 | Related Works

In the field of TSC, the optimal control problem aims to determine the most effective traffic signal timing strategies to optimize traffic flow efficiency. This requires dynamically adjusting signal timing plans based on real-time traffic conditions to reduce congestion, minimize waiting times, and ultimately enhance the overall performance of the transportation system. However, the optimal control problem is inherently complex due to challenges such as intricate intersection layouts, dynamically varying traffic flows, and multiobjective optimization requirements [9].

Historically, traffic engineers have commonly addressed traffic signal optimization by formulating the problem as an optimization model under certain traffic dynamics assumptions [13–16]. While these assumptions simplify the problem, some are unrealistic and fail to account for the complex and stochastic nature of real-world traffic. Consequently, these methods may struggle to adapt to dynamic traffic demands in a timely manner [2].

RL has recently demonstrated remarkable performance in various practical applications, including robotics, autonomous driving, unmanned aerial vehicle control, and video games. These advancements have spurred growing interest in leveraging deep RL frameworks to tackle the challenging TSC problem, yielding promising results [7, 9, 17].

However, large-scale TSC remains a significant challenge. As the number of intersections increases, the state–action space grows exponentially, complicating the extraction of relevant information and exploration of optimal policy solutions [9].

To address scalability challenges, most recent studies have adopted multiagent approaches, which can be categorized into independent multiagent reinforcement learning (MARL) and collaborative MARL. In independent MARL, agents train

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independently, whereas collaborative MARL enables coordination via mechanisms such as message passing.

Chu et al. [18] proposed the IA2C algorithm by independently applying the advantage actor–critic (A2C) algorithm to each agent. To address system instability in IA2C, they introduced two enhancements: adjacent intersection fingerprint and spatial discount factor, forming the MA2C algorithm. The former provides local policy and traffic distribution information about neighboring intersections, while the latter enables agents to focus on improving local traffic conditions.

Chen et al. [10] developed a decentralized RL model for city-scale TSC, conducting experiments in Manhattan, New York, involving 2510 traffic signals—the largest-scale experiment to date. By minimizing the "pressure" concept [19, 20], RL agents balance vehicle distribution and maximize system throughput.

The primary advantage of independent MARL lies in its scalability. However, a key limitation is that each agent observes only its local environment, which is dynamically influenced by the actions and learning processes of other agents, leading to training instability (overall nonstationarity) [9].

In contrast, collaborative MARL manages multiple intersections through coordinated agents that learn optimal policies for the entire system. Communication plays a critical role, as agents exchange information to facilitate effective coordination. Bokade et al. [21] proposed the QRC–TSC framework, which allows agents to learn communication policies to selectively transmit messages of varying lengths, minimizing communication overhead.

To enhance coordination and alleviate congestion, Yan et al. [22] developed a graph cooperation Q-learning network (GCQN–TSC) model, embedding self-attention mechanisms to enable agents to dynamically adjust their focus based on real-time traffic conditions, facilitating broader and more efficient environmental perception.

Prior works partitioned traffic networks into static regions to promote global coordination. However, static partitioning fails to adapt to dynamic traffic patterns. To address this, Ma and Wu [23] introduced a novel MARL method with adaptive network partitioning.

Due to space limitations, the relevant papers are not analyzed one by one. Table 1 presents a summary and analysis of studies on multiagent RL approaches for TSC, systematically comparing dimensions such as road network scale and type, multiagent type, RL environment (state, action, reward), algorithm, and monitoring data source.

For TSC, multiagent frameworks, while widely adopted to address scalability concerns, represent a compromise rather than the optimal paradigm due to misalignment with the problem's inherent nature. Unlike distributed scenarios such as drone swarms or warehouse AGV coordination, TSC is fundamentally a centralized optimization task—where a unified controller can perceive global network states (e.g., traffic flow, queue lengths) and coordinate all intersections holistically to mitigate systemic congestion—whereas multiagent approaches artificially decompose this task, introducing unnecessary complexity via predefined local observation scopes and interagent communication protocols. In contrast, single-agent approaches are naturally aligned with TSC's core demand, and recent advancements in RL algorithms now enable single-agent frameworks to efficiently handle high-dimensional state–action spaces in large-scale TSC. Against this backdrop, this study positions its core contribution in revisiting TSC's fundamental paradigm: we develop a single-agent RL-based method that eliminates multiagent coordination redundancy and fills the research gap of underutilizing single-agent frameworks for TSC—offering a more concise, efficient, and simple solution than mainstream multiagent approaches.

3 | Methodology

We employ single-agent RL to address TSC problems in urban corridors. RL inherently involves an agent and its environment, where the agent learns optimal strategies through continuous trial-and-error interaction with the environment, eliminating the need for prior knowledge of the environment's dynamic model.

All notations are summarized in Table 2.

During the learning phase, in each iteration, the agent observes the environment state at time step t, denoted as S_t. This state serves as a comprehensive representation of the environment, encapsulating all necessary information for the agent to take optimal actions. The agent then executes an action a_t based on S_t to interact with the environment. Consequent to the current state S_t and selected action a_t, the environment transitions to a new state S_{t+1}, and the agent receives the new state S_{t+1} and a reward signal r_{t+1} from the environment, which quantifies the outcome of the action a_t. Leveraging these trial-and-error iterations and an optimal strategy learning algorithm, the agent iteratively refines its action–selection strategy to achieve specific control goals.

The iterative process of applying RL to TSC is illustrated in Figure 1. The above framework highlights that RL-based TSC necessitates careful attention to the following core components: state definition, action design, reward function formulation, environment, and strategy learning algorithm.

It should be clarified that during the learning phase, the state can be obtained from the vehicle travel data in the simulation model, without the need for actual probe vehicle data. However, during the deployment phase of the trained model, the state needs to be estimated in real time using actual probe vehicle data. Thus, the traffic state indicators involved in the state design must be estimable using probe vehicle data. It is also important to note that the reward is not needed during the model deployment phase, and thus, the reward design is not constrained by this requirement.

3.1 | State

The state consists of two parts: the current congestion state and the current signal phase scheme.

3.1.1 | Current Congestion State

The congestion state is represented by the queue length for each link (segment) at the current time. The congestion state is a L vector, where L represents the number of links in a corridor:

$$[q_t^l], \tag{1}$$

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TABLE 1 | Overview of multiagent RL approaches for TSC.

Literature	Scenario	Type of MARL	Status	Action	Reward	Algorithm	Data source	Other
Chu et al. [18]	5 × 5 grid network; a real-world network with 30 intersections	Independent MARL	delay; total number of approaching vehicles	Select phase set [24]	Queue; delay	MA2C	Induction-loop detectors (ILD)	LSTM
Chen et al. [10]	A real-world network with 2510 intersections	Independent MARL	The current phase; the pressure of the 12 traffic movements	Select phase	The pressure of intersection *i*	FRAP algorithm [25]	Probe vehicle	Addressing coordinated control based on the "pressure" mechanism [19, 20]
Agafonov and Myasnikov [26]	6 × 6 grid network; a real-world network with 196 intersections	Independent MARL	Current traffic signal phase; queue length on each incoming lane; number of vehicles on each spatial segment of the incoming and outgoing lanes	Select phase	Queue length; average speed; pressure	IDQN	Probe vehicle	
Wang et al. [27]	5 × 5 grid network; a real-world network with 30 intersections	Independent MARL	The number of vehicles at intersection; the cumulative wait time of all vehicles at intersection	Select phase	The queue length at each incoming lane; the cumulative wait time of all vehicles at intersection	Region-aware cooperative strategy (RACS) based on graph attention network (GAT [28]) upon IA2C [29]	Induction-loop detector	Graph attention network (GAT) was introduced to address some observability issues in the IA2C system.
Yan et al. [22]	A real-world network with 48 intersections	Collaborative MARL	Queue length; waiting time; delay; current phase; duration of the current phase	Select phase	Queue; waiting time; CO; NO	Graph cooperation Q-learning network traffic signal control (GCQN–TSC)	Camera	

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TABLE 1 | (Continued)

Literature	Scenario	Type of MARL	Status	Action	Reward	Algorithm	Data source	Other
Bokade et al. [21]	4 × 4 grid network; a real-world network with 7 intersections	Collaborative MARL	The number of vehicles; the average normalized speed of the vehicles; the number of halted vehicles (queue lengths); the current phase ID of the traffic signal	Select phase	Queue length	QRC–TSC upon the existing QMIX [30] and NDQ [31] frameworks	Lane area detector	Enables agents to selectively choose the recipients of their messages and exchange variable-length messages with them.
Ma and Wu [23]	4 × 4 grid network; three real-world road networks, consisting of 12, 16, and 48 intersections, respectively.	Collaborative MARL	Queue length	Select phase	Queue length	A variant of QMIX	Probe vehicle	Adaptive network partition.
Zhu et al. [32]	A real-world network with 16 intersections	Collaborative MARL	The number of queuing vehicles	Extend the current phase or switch to a different one	The predicted number of queuing vehicles	Max-plus algorithm	Loop detectors	A coordination graph-based framework
Fu et al. [33]	5 × 5 grid network; two real-world road networks consisting of 12 and 16 intersections, respectively.	Collaborative MARL	Queue length	Select phase	Queue length	Contrastive learning (CL)–based MARL	Probe vehicle	

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TABLE 2 | Table of notations.

Notations	Explanation
a	The action of the agent
S	The state observed by the agent
r	The reward obtained by the agent
q_t^l	The queue length on the link $l \in L$ at time t
q_{ub}	The upper bound of the queue length
L	The number of links in the corridor
s_t^m	The signal phase split at the intersection $m \in M$ at time t
s_{lb} and s_{ub}	The lower and upper bounds of the signal phase split
Δs	The phase split adjustment magnitude
M	The number of signal intersections in the corridor
q_{lc} and q_{hc}	The thresholds for light and heavy congestion, respectively
w_l	The link importance weight
w_{cp}	The penalty weight associated with heavy congestion

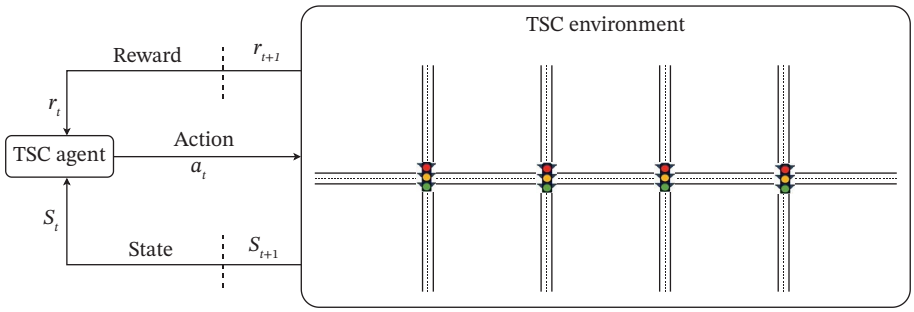


FIGURE 1 | Single-agent RL architecture for corridor traffic signal control.

where q_t^l is the queue length on the link $l \in L$ at time t. The queue length is bounded by 0 and q_{ub}, which are predetermined.

Figure 2 illustrates the definition of queue length in this paper. The figure presents the spatiotemporal trajectories of a vehicle platoon that enters a specific road link and travels straight through both upstream and downstream intersections within the same green light cycle (hereinafter referred to as the "through fleet"), and also depicts the queued vehicles at the downstream intersection. The queue length in this paper is defined as the number of queued vehicles. This queue length has a significant correlation with the average travel time of the through fleet: Specifically, as queued vehicles take up green light time, longer queue lengths lead to more vehicles in the through fleet encountering red lights, thereby increasing the average travel time. Thus, queue length serves as a highly relevant indicator of road link congestion.

The reason for adopting the above definition of queue length is primarily that it can be estimated using probe vehicle data. Previous studies proposed methods to estimate average link travel time [34] and queue length [12] using probe vehicle data, respectively. Due to space limitations, this paper does not reiterate these methods. In these methods, the probe vehicles represent a small subset of the through fleet. The data utilized consist solely of link travel times provided by these probe vehicles, without directly incorporating raw data such as trajectory points.

3.1.2 | Current Signal Phase Scheme

The current signal phase scheme is represented by a signal phase split for each intersection of the corridor. This is a M vector, where M represents the number of signal intersections in the corridor:

$$[s_t^m], \tag{2}$$

where s_t^m represents the signal phase split at the intersection $m \in M$ at time t. The signal phase split is bounded by s_{lb} and s_{ub}, and these values are predetermined.

In this study, each intersection is controlled by a four-phase signal (see Figure 3). It is assumed that the signal cycle, left-turn phase time, yellow time, all-red time, and offset time are predetermined and remain constant. The signal phase split is the only adjustable variable and is defined as the sum of the north–south phases' time, which includes the north–south straight/right-turn phase (Phase 1) and the left-turn phase (Phase 2). Since the signal cycle is constant, adjusting the signal phase split can control traffic flow at upstream and downstream intersections, thereby managing congestion levels. As illustrated in Figure 3, if congestion occurs on the link indicated by the arrow (from intersection m to $m+1$), the congestion can be alleviated by reducing the number of vehicles entering this link by increasing the signal phase split at the upstream intersection (intersection m), and/or by increasing the number of vehicles

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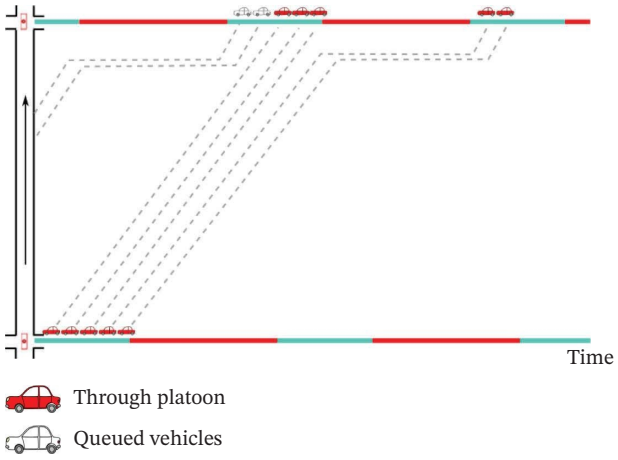


FIGURE 2 | The definition of queue length.

exiting this link by decreasing the signal phase split at the downstream intersection (intersection $m+1$).

The above definition involves certain simplifications. The most important issue is that it does not account for the regulation of left-turn traffic flow. Since left turns constitute a very small proportion of overall trips, it is reasonable to overlook this part. Notably, reducing the adjustment of an entire intersection's timing plan to a single variable (phase split) can significantly reduce the state dimension, making it a critical foundation for enabling effective policy learning in the single-agent framework.

3.2 | Action

The action is divided into two stages: first, selecting an intersection, and then adjusting the phase split of that intersection.

For an intersection, the agent has three possible actions: adjusting the current signal phase split by $-\Delta s$, 0, and Δs, respectively, where Δs is a predefined value.

It should be noted that this adjustment is cumulative, and the phase split can ultimately be adjusted to the preset lower and upper bounds (s_{lb} and s_{ub}).

This action design can directly adjust the traffic flow at intersections and alleviate traffic flow on road segments. As shown in Figure 3, if congestion occurs on the road segment indicated by the arrow (from intersection m to $m+1$), the congestion on this segment can be alleviated by either selecting intersection m and increasing the phase split by Δs to reduce the number of incoming vehicles, or selecting intersection $m+1$ and decreasing the phase split by Δs to increase the number of outgoing vehicles.

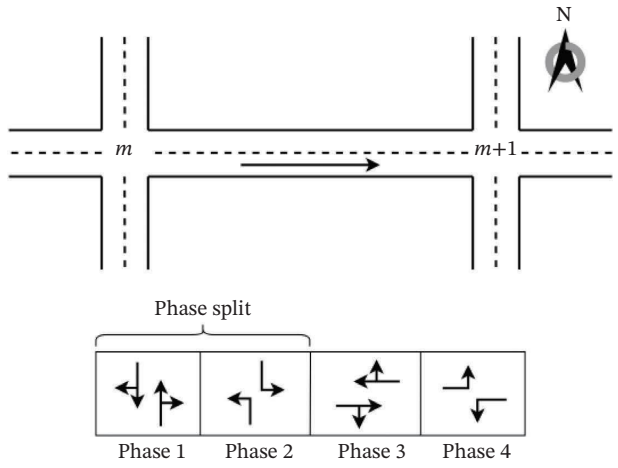


FIGURE 3 | The signal phase split.

The action design can also significantly reduce the size of the action space, which is also the key to realizing multi-intersection control using a single-agent framework. If there are M intersections, the size of the action space is $M \times 3$. It is worth noting that this design ensures that as the number of intersections increases, the action space will not experience explosive growth. In contrast, if the timing of all intersections is adjusted simultaneously, the corresponding action space would be $3^{\wedge}M$, which would lead to an explosive expansion of the action space.

3.3 | Reward

The queue length is selected as the indicator of congestion level, and the reward is formulated as a function of the queue length of each link in a corridor. Additionally, a penalty weight is incorporated in cases of severe congestion to achieve the goal of this study, which is to alleviate congestion.

The reward for a corridor is calculated as the sum of the rewards of its constituent links, with the reward for an individual link defined as shown in the following equation:

$$\begin{cases} q \le q_{lc}\,(\text{Free flow}): \ \text{reward} = 0, \\ q_{lc} \le q \le q_{hc}\,(\text{Light congestion}): \ \text{reward} = -(w_l \times q), \\ q \ge q_{hc}\,(\text{Heavy congestion}): \quad \text{reward} = -(w_{cp} \times w_l \times q), \end{cases} \tag{3}$$

where q is the queue length of a link and q_{lc} and q_{hc} are the thresholds for light and heavy congestion, respectively. w_l represents the link importance weight. w_{cp} is a large penalty weight associated with heavy congestion, aiming to use the significant negative reward to minimize the occurrence of such states as much as possible. The values of q_{lc}, q_{hc}, w_l, and w_{cp} are constants determined in advance.

When the queue length is below q_{lc}, it is regarded as a state close to free flow and is preferable, so there is no negative reward. When the queue length is between q_{lc} and q_{hc}, it falls into a slow-moving state that is tolerable, with a reward equal to $-q$ and no additional penalty coefficient applied. When the queue length exceeds q_{hc}, it is considered a relatively severe congestion state and is undesirable; a larger penalty coefficient is added here, aiming to minimize the occurrence of such a state as much as possible.

3.4 | Environment

We develop the TSC environment using Gymnasium [35], an open-source Python library designed for the development and comparison of RL algorithms. Gymnasium facilitates seamless interaction between learning algorithms and environments through its standardized API.

For the observation space, we concatenate the queue lengths and phase splits into a vector of length $(L+M)$, then normalize each

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element to the interval [0, 1] using its respective lower and upper bounds.

For the action space, we discretize all possible actions into integers, resulting in a total of $3 \times M$ integers.

The core components of the TSC environment consist of a traffic corridor simulation model, a simulation controller, and a traffic monitor. The simulation model should closely mirror the real-world traffic corridor targeted for signal timing optimization, encompassing road geometries, intersection geometries, and traffic demand patterns. The simulation controller is responsible for managing the simulator's operations, including running the simulation model and dispatching signal control schemes to the simulation model.

The traffic monitor is responsible for collecting the queue lengths of all road segments in the traffic corridor. In this paper, queue lengths are estimated using data generated by all vehicles within the simulation platform, rather than through probe vehicle technology. This approach is valid during the RL strategy's learning stage. However, during model evaluation and deployment, probe vehicle technology should be adopted. To simplify the evaluation process, this paper uses the same approach employed in the training stage to estimate queue lengths. Since queue length estimation based on probe vehicle data is a solved problem, this simplification has minimal impact on the model's evaluation.

At a specific control time (t), the queued vehicles on a given road segment consist of two parts, and the queue length of the road segment can be obtained through these criteria (see Figure 4).

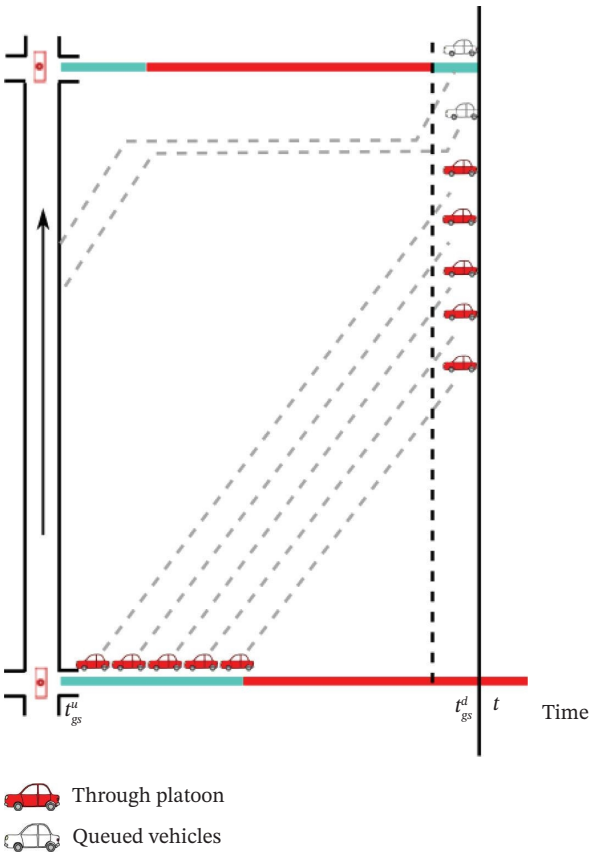


FIGURE 4 | The queued vehicles on a road segment.

1. Vehicles passing through this segment shall meet the following criteria: proceeding straight through the downstream intersection, with an entry time before the start time of the most recent green phase at the upstream intersection (t^u_{gs}) and a departure time after the start time of the next green phase at the downstream intersection (t^d_{gs});
2. Vehicles remaining on this segment shall meet the following criteria: intended to proceed straight through the downstream intersection and with entry time before the start time of the most recent green phase at the upstream intersection (t^u_{gs}).

3.5 | Strategy Learning Algorithm

We did not develop a custom RL learning algorithm; instead, we utilized RLlib [36] to train the proposed TSC model. RLlib, an open-source RL library, supports production-level, scalable, and fault-tolerant RL tasks while offering unified APIs for diverse applications.

The key advantages of using RLlib are twofold. First, it allows for straightforward training of Gymnasium-compliant RL environments with minimal configuration, benefiting traffic engineering researchers who may lack expertise in algorithm development. Second, RLlib enables the evaluation and comparison of multiple state-of-the-art (SOTA) algorithms (e.g., PPO, DQN) within a standardized framework. This feature is crucial for benchmarking algorithm performance across different traffic scenarios, facilitating the identification of optimal strategies without extensive debugging.

By adopting the RLlib framework, researchers only need to focus on the development of the TSC environment. The RLlib framework, as a mature RL toolkit, provides built-in modules for algorithm training, parameter tuning, performance evaluation, and a detailed and comprehensive development document—this not only reduces the repetitive development workload for researchers but also ensures the stability and reproducibility of the control strategy learning process.

4 | Experiment Design

4.1 | Simulation Setup

The experiments are conducted in a simulation environment using SUMO, a sophisticated and adaptable traffic simulation software. SUMO features a dedicated interface known as libsumo, which allows users to manipulate the simulation through Python scripts. By leveraging the libsumo interface, the simulation controller can control simulation running, access real-time traffic data, and dispatch signal schemes to SUMO. The mesoscopic mode is used, and it is 100 times faster than the microscopic mode.

The RL-based TSC method is tested within an urban corridor, and the geometry of this corridor is depicted in Figure 5. Each link has three lanes and extends to four lanes at the downstream intersection. Each approach consists of two straight lanes, one left-turn lane and one right-turn lane. Each intersection is controlled by four signal phases, with their sequence predetermined. The signal cycle, left-turn phase time, yellow time, and all-red time remain constant and are set to 100, 8, 2, and 2 s, respectively. The initial signal phase split is set to 50 s.

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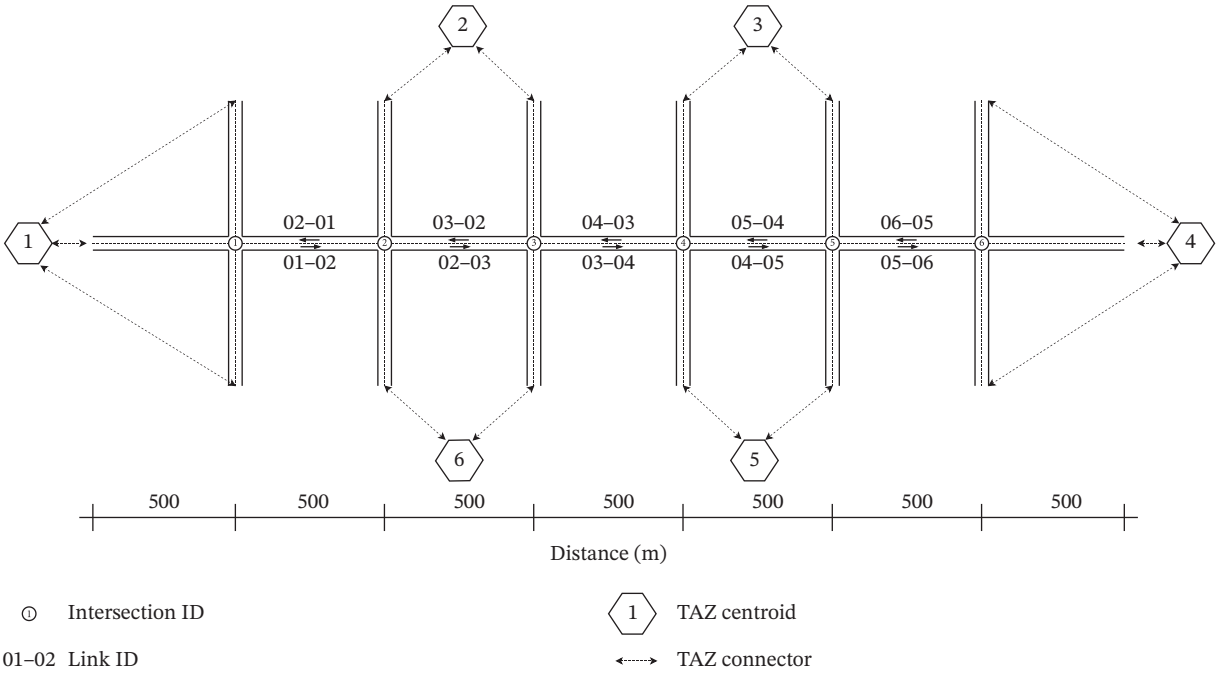


FIGURE 5 | Corridor diagram.

Traffic is generated using zone-to-zone demand generation, where traffic assignment zones (TAZs) and connectors are depicted in Figure 5. Zone-to-zone flows are defined by an OD matrix. Two patterns of OD matrix scenarios are tested. In the first pattern of the OD matrix (Scenario 1), the west-eastbound flow is set to be dominant to simulate morning peak traffic in a corridor connecting the suburban and central areas of a city. In the second pattern (Scenario 2), the east–west and west–east bound flows are set to be equal, simulating morning peak traffic in a corridor located in the central area of a city.

4.2 | RL-Model Parameter Setting

Table 3 summarizes the parameters of the RL-based TSC model. When the signal phase split is set to its initial value (50 s), the saturated flow of the straight lanes within a signal cycle is estimated to be 50 vehicles. When the queue length reaches this threshold (50 vehicles), the queued vehicles occupy the entire green time, and all new incoming vehicles will encounter a red light (as illustrated in Figure 2). In this paper, such a state is considered highly congested and strictly undesirable; therefore, the upper bound of the queue length (q_{ub}) is set to this value. The thresholds for light and heavy congestion (q_{lc} and q_{hc}) are set at 20% and 50% of this saturated flow level. These thresholds are used to classify free flow, slow-moving, and congested states. Currently, there are no universally recognized standards for them, and they can be set based on engineers' experience in practical scenarios.

The upper and lower bounds of the signal split (s_{lb} and s_{ub}) are set to 30 and 70 s, respectively, meaning there is an adjustment range of ± 20 s relative to the initial value. In this experiment, the setting of these values is somewhat arbitrary. In practical scenarios, a reasonable range can be established under the constraint of the minimum green light duration. The adjustment step size (Δs) is set to 2 s. This setting also has some arbitrariness, but in practical scenarios, reasonable values can be tested within the range of 2–5 s. In Scenario 1, where the major traffic flow is from west to east, the link importance weights (w_l) are set to 0 for the links in the opposite direction. This is done to consider only the links that are consistent with the major traffic flow in the reward calculation.

The RL policy network consists of two fully connected layers, each containing 32 neurons.

The RL models are trained over multiple episodes, with each episode simulated for 16,200 s, including a 1800-s warm-up period. During the warm-up, the signal scheme is unchanged, and the vehicles are incrementally introduced into the road network to establish a stable initial state. Each episode consists of 144 steps, calculated by dividing the remaining time after the warm-up period (16,200–1800 s) by the control interval (t_c).

RLlib library is used to learn a signal control policy. This study investigates PPO and DQN, using the default hyperparameter configurations from RLlib.

In this study, we employed a computing platform equipped with an Intel i9-13900 K processor (24 physical cores and 32 threads) and an NVIDIA GeForce GTX 4090 graphics card.

5 | Simulation Results and Discussions

5.1 | Comparison of RL Algorithms

As illustrated in Figures 6 and 7, the PPO algorithm demonstrates significantly superior convergence efficiency over the DQN across both traffic scenarios. Specifically, PPO achieves stable high episode rewards within 1 h in both Scenario 1 and Scenario 2, with its convergence curve showing a monotonically increasing trend and a significantly smaller fluctuation amplitude than that of DQN. This characteristic is attributed to PPO's on-policy mechanism, which directly utilizes data generated by the current policy's interaction with the environment for training,

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TABLE 3 | RL-based TSC method parameters.

Parameter	Value
q_{ub}	50 no. of vehicles
q_{lc} and q_{hc}	10 and 25 no. of vehicles (20% and 50% of q_{ub}, respectively)
s_{lb} and s_{ub}	30 and 70 s
Δs	2 s
w_l	0 for the east–west bound links in the first pattern of the OD matrix; 1 for otherwise
w_{cp}	10

enabling more precise adaptation to the dynamic changes in traffic flow. In contrast, as an off-policy algorithm, DQN relies on an experience replay buffer to enhance data efficiency but struggles to handle the environmental nonstationarity in the highly complex state space of traffic control, leading to severe oscillations or even degradation in reward values during training.

The superior performance of PPO in this study confirms the unique value of on-policy algorithms in dynamic optimization problems. TSC inherently belongs to a nonstationary Markov decision process (MDP) with continuous state spaces, where the state transition matrix evolves in real-time with traffic flow distribution. By integrating generalized advantage estimation (GAE) and policy clipping, PPO ensures the safety of policy updates while enabling rapid adaptation to nonstationary environments. In contrast, DQN's experience replay mechanism suffers from sample sparsity in large state spaces and struggles to handle the state transition correlations prevalent in traffic networks (e.g., spillover effects between adjacent intersections), further exacerbating the learning difficulty.

In summary, experimental results clearly demonstrate that PPO is better suited as the underlying optimization algorithm for urban corridor TSC. Its fast convergence and environmental robustness provide critical technical support for real-time deployment in practical scenarios.

5.2 | Performance of TSC

The performance of the TSC model is compared against the baseline case, which involves no adjustments to the signal timing schema, across both Scenario 1 and Scenario 2. The TSC model was trained using the PPO algorithm, with each of the two scenarios subjected to a training duration of 12 h.

As illustrated in Figures 8(a) and 9(a), the baseline case in Scenario one exhibits severe queue length violations. Figure 8 depicts the changes in queue length at each link throughout an episode for Scenario 1. For Scenario 1, only the queue lengths for links aligned with the main traffic flow direction are shown. In Figure 8, the upper limit of queue length is set to 50. However, the actual queue length in the baseline case far exceeds this limit. Figure 9 illustrates the distribution of queue lengths for each road segment for Scenario 1. The total number of queue length data points equals the product of the number of road segments and the number of signal cycles. Specifically, for Scenario 1, this is calculated as $5 \times 144 = 720$. As depicted in the figures, under the base case, the queue length exceeded 50 on multiple occasions and even surpassed 150.

As shown in Figures 8(b) and 9(b), the proposed TSC model demonstrates remarkable effectiveness in alleviating urban corridor congestion when compared against the baseline case. In Figure 9, for the controlled case, to eliminate randomness and ensure the reliability of the results, five episodes were evaluated. As shown in these figures, the queue length did not exceed 50, and situations where it exceeded 30 were also relatively few. That is, the results demonstrate that the approach significantly reduces queue lengths and improves overall traffic flow efficiency across the evaluated scenarios.

For Scenario 2, Figures 10 and 11 show results similar to those in Scenario 1. In Scenario 2, simulating balanced bidirectional traffic in urban centers, the TSC model demonstrates consistent performance across all links. Notably, the model achieves

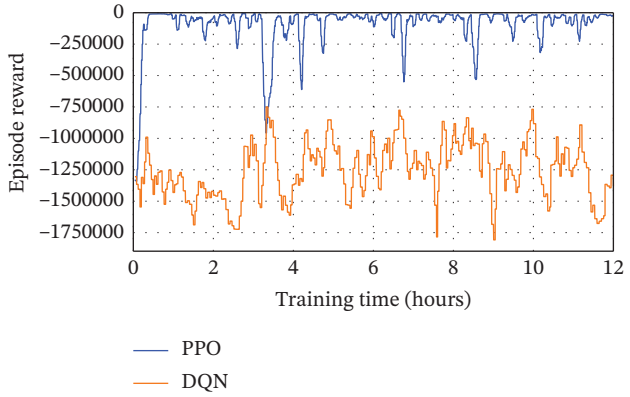


FIGURE 6 | Learning process (Scenario 1).

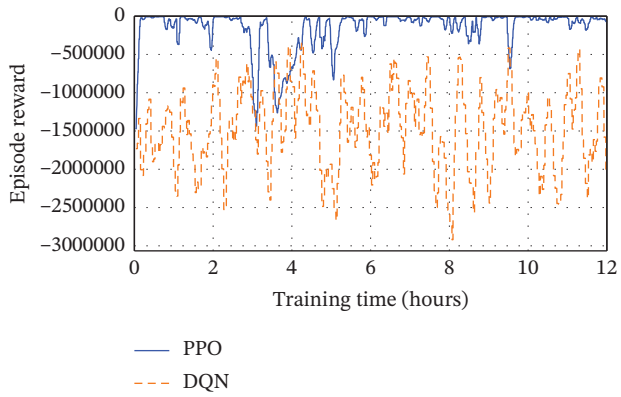


FIGURE 7 | Learning process (Scenario 2).

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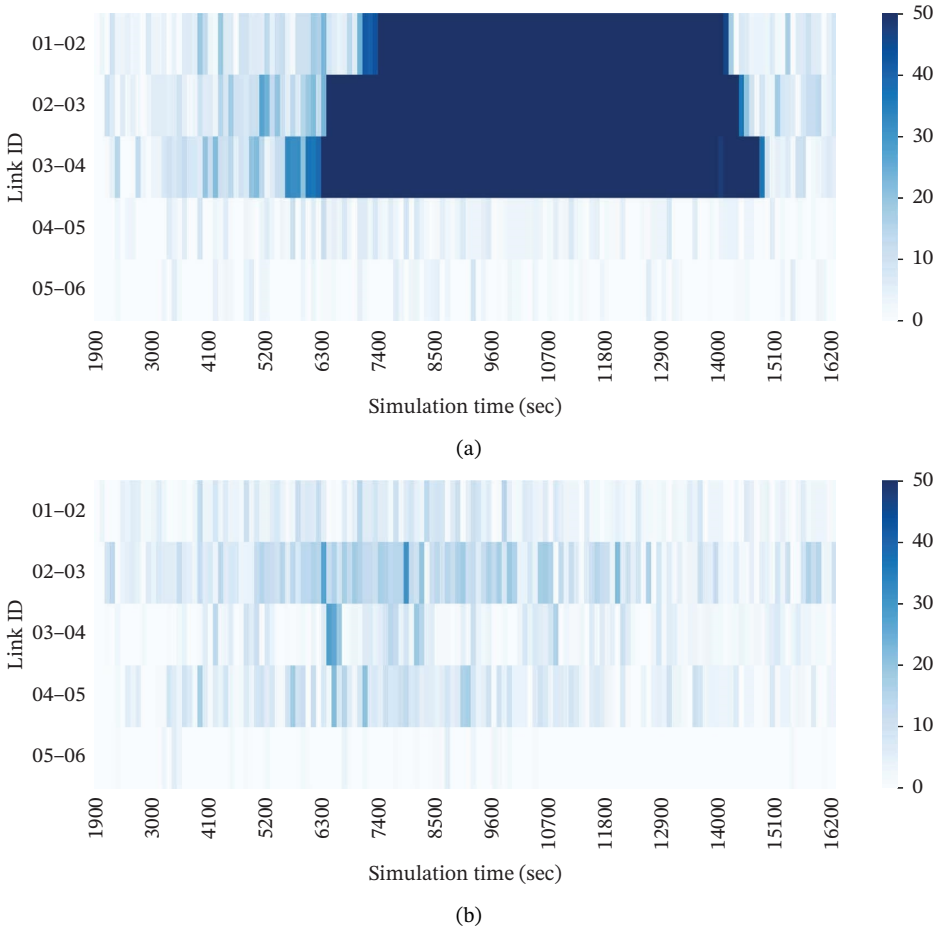


FIGURE 8 | Queue length comparisons in Scenario 1. (a) Baseline case and (b) controlled case.

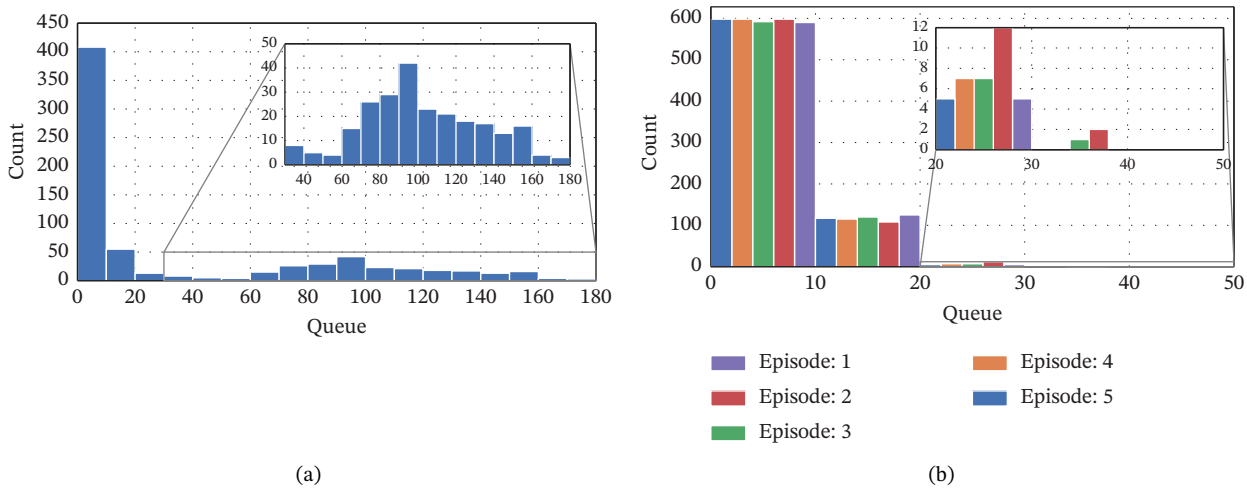


FIGURE 9 | Queue length distributions in Scenario 1. (a) Baseline case and (b) controlled case.

symmetric congestion control for both east–west and west–east flows, with queue lengths stabilized within 20–35 vehicles across all segments. This contrasts with the baseline case, where alternating congestion waves occur due to fixed signal timings, leading to queue lengths fluctuating between 50 and 160 vehicles. The standard deviation of queue lengths in the controlled case is

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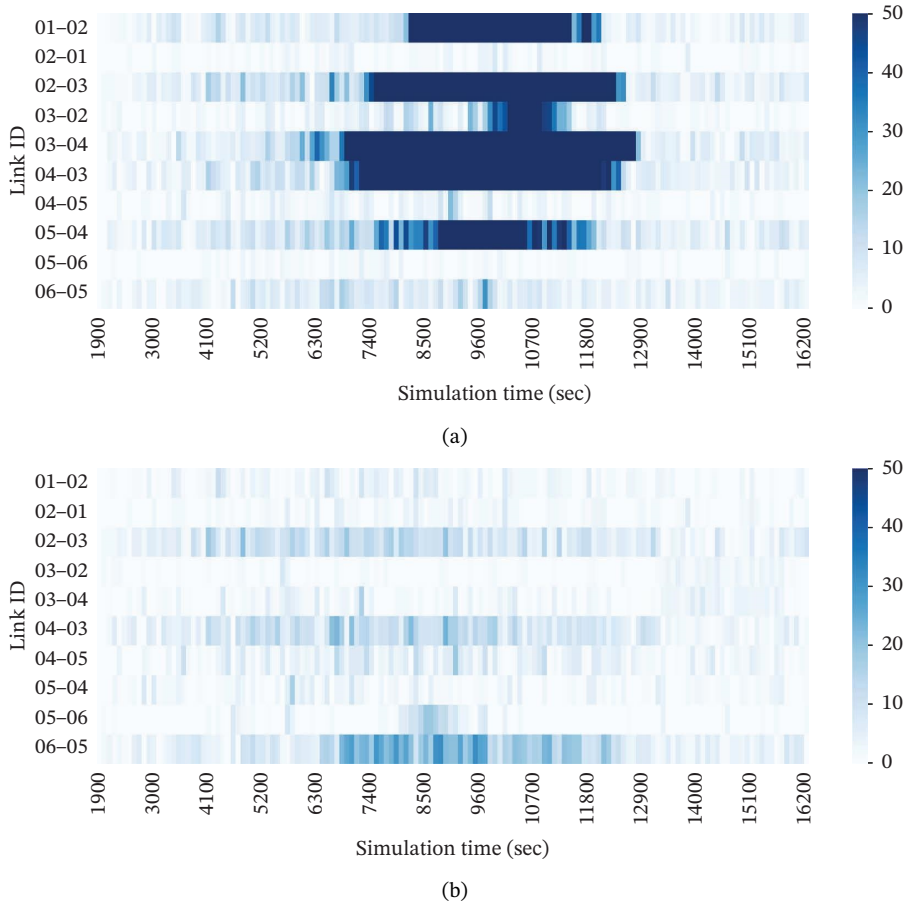


FIGURE 10 | Queue length comparisons in Scenario 2. (a) Baseline case and (b) controlled case.

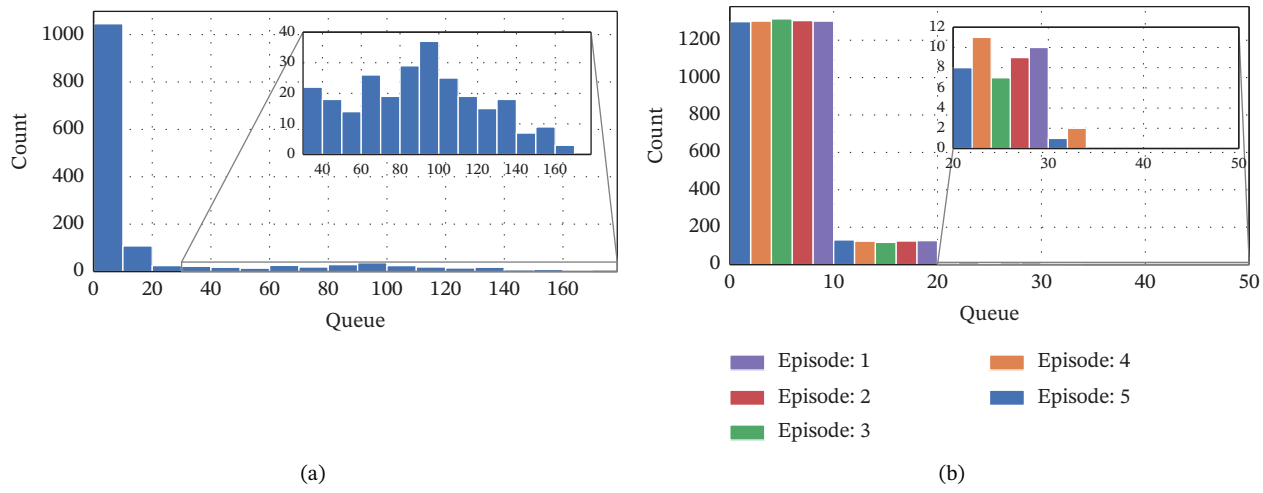


FIGURE 11 | Queue length distributions in Scenario 2. (a) Baseline case and (b) controlled case.

reduced by 68% compared to the baseline, indicating significantly improved traffic flow stability.

In summary, the TSC model demonstrates robust congestion mitigation capabilities across diverse traffic patterns, leveraging real-time queue length estimation and adaptive signal control to create a more balanced and efficient urban transportation system.

6 | Conclusion

This study endeavors to address multi-intersection control in adaptive traffic signal control (ATSC) via a single-agent RL framework, departing from the dominant multiagent paradigm in existing research. The core of this innovative framework lies in the rational design of state, action, and reward.

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First, efforts were focused on reducing state and action space dimensionality, a key prerequisite for single-agent policy learning. State design simplified an intersection's signal timing scheme to one adjustable variable: signal phase split, retaining the parameter governing traffic flow while eliminating redundancy. Action design adopted a two-step mechanism: select an intersection first and then adjust its phase split in three directions by a predefined step. This limits the action space to $M \times 3$ (M = number of intersections), avoiding exponential expansion (3^M if adjusting all intersections simultaneously) and enabling multi-intersection scalability.

Second, state, action, and reward were tightly coupled with queue length (a direct congestion indicator). The state integrates real-time link queue lengths (spatial congestion) and intersection phase splits (control status); actions adjust phase splits to modulate traffic flow and queue dynamics; rewards impose hierarchical penalties based on queue length. This strong correlation forms a clear feedback loop, accelerating policy convergence by helping the agent quickly learn action-congestion mitigation mappings.

Notably, the design involves simplifications (e.g., ignoring left-turn flow regulation, using phase split as the sole adjustable parameter). However, experiments validate its effectiveness: the PPO algorithm achieves stable convergence within 1 h, and the model eases corridor congestion—restricting queue lengths to ≤ 50 vehicles (predefined upper bound), versus baseline fixed timings where queues often exceeded 100 vehicles.

Another key advantage is the model's practical deployability. Its state relies on queue length, estimable via prevalent probe vehicle data (e.g., ride-hailing/navigation vehicles). Probe vehicles cover most urban roads without requiring additional infrastructure, enabling rapid real-world deployment. Since queue length estimation via probe data is a well-established technique, this study did not revalidate it; future work can integrate real probe data to test performance in actual traffic.

In summary, this study demonstrates the feasibility and superiority of single-agent RL for multi-intersection ATSC, offering a practical solution for corridor congestion. Future research will extend the model to large-scale networks (hundreds of intersections) to test scalability and explore advanced RL algorithms for adaptability to complex traffic. Based on this research, a related study [37] investigated a single-agent RL method for regional traffic control and validated its effectiveness using a simulation model of 25 grid-style intersections. However, this preprint is currently in a withdrawn state, as the authors deemed the research to be incomplete and not yet perfected.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available upon reasonable request from the author, Qiang Li (liqiangjl@qq.com).

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